

Essentials of Data Science

Introduction to Data Science
CS5113

Course Instructor : Dr. Nidhi Goyal

8/6/2026

Who am I?

- **Research interests**

- Generative AI, Large Language Models, Natural Language Processing, Knowledge Graphs, Graph Neural Networks, Computational Neuroscience, Computer Vision, Brain Computer Interfaces, Generative AI and solving real-world problems

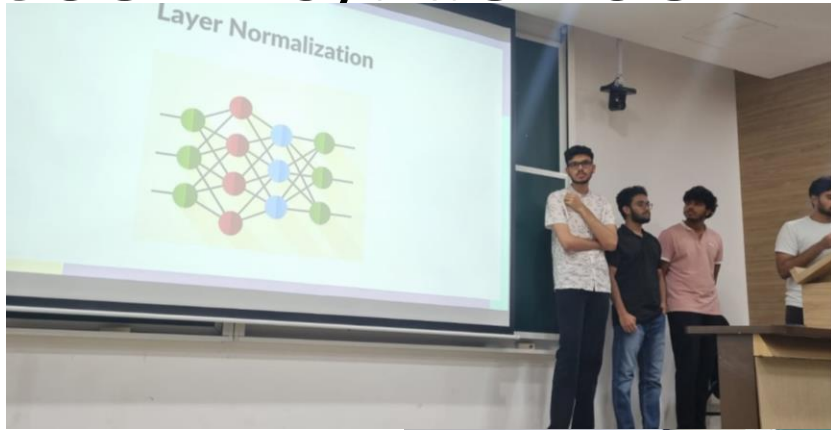
- **Experience (Research and Teaching)**

- Taught Generative AI & LLM (CS5202), NLP for multiple semesters (CS3216/AI5203) to B.Tech (CSE/AI/ECM) & M.Tech students.
- Ph.D from IIIT-Delhi and PMFDR fellow in collaboration with InfoEdge India Limited
- Applied researcher in NLP, Knowledge Graphs, Graph Neural Networks
Former Assistant Professor, GGSIPU, New Delhi
- Former Assistant Professor, DCE, Gurgaon
- TA for many courses- COA, Privacy and security on Online social media @IIIT, NPTEL, NLP course PGDSAAI@IIIT-Delhi



Dr. Nidhi Goyal
(Assistant Professor,
Ecole School of
Engineering)

Previous Course Offerings of other courses (NLP) /CS3216/AI5203



Lecture Plan

- Course Logistics
- Course Outcomes
- Content/Syllabus
- Grading Policy
- Why should you learn Essentials of Data Science?
- What is Data?
- Applications
- Resources

Course Logistics

- Attendance is mandatory!!
- All attendance is via QR codes
- *"Missing Minor exams will not be given re-test. If a student misses it due to genuine and verifiable reason(s), she/he will be given weightage for the missed exam(s) on a pro-rate basis of performance in the end-sem exam."*

- Lecture Timings [**ELT-1**]:
 - **Monday: 9:25-10:20 AM**
 - **Wednesdays- 9:25-10:20 AM**
 - **Fridays- 10:25 AM–11:30 AM** (Strictly adhere to class timings)
- Slack, help sessions/office hours (for all course questions/discussion)
 - Slack for assignments, doubts, course announcements, deadlines, etc.
 - Office hours start **Wednesday (4:00-5:00 pm)**, Location- **IT2 block, 2nd floor, New faculty office!**
- For any queries: nidhi.goyal@mahindrauniversity.edu.in
 - Use email judiciously for important/urgent communication
- Follow Course page for resources, materials, etc. We've put a lot of other important information on the class webpage (Link to be published soon).
- Slide PDFs and other resources shall be uploaded before each lecture.

Smooth functioning of Course

- Slack for assignments, doubts, course announcements, deadlines, etc.
 - Create your account on slack with **Your Roll number as username**.
 - Get yourself added to below channels by your instructor or TAs
 - **course-announcements**
 - Be active to check slack notifications for course related information!!
 - First communication point for doubts on slack (**#doubts** channel).
 - Healthy discussions among peers lead to effective learning.

You are expected to join the course Slack workspace, and are responsible for monitoring it for any announcements posted by the instructor or the TAs - all such information constitutes official course communication.

Slack workspace

https://join.slack.com/t/eds-wze2171/shared_invite/zt-45z6n7w01-e87YUUCf~7c3tkq6cB7BKA



Course Outcomes

After the successful completion of this course , the students will be able to:

- Acquire and process tabular, textual, or hierarchical data.
- Uncover patterns by summarizing and visualizing data.
- Apply machine learning to answer real-world prediction problems.

Pre-requisites

- Math, Probability, Basic knowledge of Python, Statistical concepts

Course Content/Syllabus

Details are attached.

Course Web Page. The course web page is on [Teachings](#). The course web page will contain all materials distributed in class either in hardcopy or electronically, including all lecture notes, assignments and Jupyter/Colab notebooks.

Don't mistake this class for a easy to go version of Data Science.

You will be prepared for most “Data Scientist” and “Data Analyst” internships after taking this class.

Grading Policy

Type of Evaluation	% Contribution in Grade
Midsem	25
Continuous assessments (Quiz/Assignments)	20
Project (Implementation 20% + Report 5% + Presentation/Viva 5%)	30
Endsem	25

Cheating Policy

<https://www.mahindrauniversity.edu.in/sites/default/files/2022/Annexure-XVI-Unfair-Means.pdf>

- Exams and quizzes are to be completed individually. Verbal collaboration on homework assignments is acceptable, but
 - you must not share any code or other written material,
 - Everything you submit must be your own work, and
 - you must note the names of anyone you collaborated with on each problem (the only exceptions are the instructors and TAs), and the nature of the collaboration (e.g., “X helped me,” “I helped X,” “X and I worked it out together.”).
- If you find material in published literature (e.g., on the Web) that is helpful in solving a problem, you must cite it and explain the answer in your own words.
- The project is to be completed by a team; you are not permitted to discuss any aspect of your project with anyone other than your team members, the instructor, and the TAs.
- You are encouraged to use existing NLP components in your project; you must acknowledge these appropriately in the documentation.



The influence of technology on academic distraction: A review

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Abstract

The detrimental effects of academic distraction have become increasingly problematic in recent years due to student media multitasking. In recent years, educators have gradually transitioned many classes to the online learning environment, effectively reducing teacher supervision. This has been greatly exacerbated with the advent of coronavirus disease 2019 and subsequent social distancing guidelines. In an effort to help students and educators better understand the implications of technology on academic distraction and academic performance, this literature review discusses the effects of three forms on technology on student distraction: laptops, smartphones, and social media use, particularly the social media platform Facebook. While the effects of student laptops on academic distraction are somewhat ambivalent, the debate is largely moot in the current educational environment. The detrimental effects of student smartphone and social media use on academic distraction are more conspicuous, especially with the pervasiveness of personal digital devices. Potential self-regulatory and environmental interventions are discussed.

KEYWORDS

academic distraction, Facebook, laptops, learning, media multitasking, online learning, performance, phubbing, smartphones, social media



No Mobile phones during the class!!

Academic Performance Degradation

A study by the National Center for Education Statistics revealed that students who engaged with their phones during class sessions scored an average of **11% lower** on tests and assignments.

The underlying mechanisms responsible for this performance degradation include:

- Reduced Attention Span:** The constant stream of notifications and digital stimuli primes the brain for instant gratification, diminishing the capacity for sustained focus on complex academic material.
- Working Memory Overload:** Frequent task-switching between academic content and digital distractions can overload working memory, hindering the encoding of new information into long-term memory.
- Context Switching Costs:** Shifting focus between different tasks, even momentarily, incurs a cognitive cost. Re-engaging with the primary task requires additional processing power, further diminishing overall efficiency.

Mobile phone Replacement with Neurobics (Some mental exercises for 2-3 mins!)



Data Data Everywhere

In today's world, especially here in India, **we're surrounded by data**. Every day, we **generate, transmit**, and often **give away** data without realizing it. Each click, scroll, tap, text, or voice note can become a variable influencing business decisions and economies.

[Current World Population: 8,298,979,488](#)

OCT
2025

POPULATION ESSENTIALS

DEMOGRAPHICS AND OTHER KEY INDICATORS



TOTAL
POPULATION



8.25
BILLION

FEMALE
POPULATION



49.7%

MALE
POPULATION



50.3%

YEAR-ON-YEAR CHANGE
IN TOTAL POPULATION



+0.8%
+69 MILLION

MEDIAN AGE OF
THE POPULATION



30.9

URBAN
POPULATION



58.4%

POPULATION DENSITY
(PEOPLE PER KM²)



63.1

OVERALL LITERACY
(ADULTS AGED 15+)



87.4%

FEMALE LITERACY
(ADULTS AGED 15+)



84.1%

MALE LITERACY
(ADULTS AGED 15+)



90.6%

15

SOURCES: KEPIOS ANALYSIS; UNITED NATIONS; LOCAL GOVERNMENT AUTHORITIES; WORLD BANK; UNESCO; CIA WORLD FACTBOOK; OUR WORLD IN DATA; INDEXMUNDI; KNOEMA. ALL VALUES REFLECT THE LATEST AVAILABLE DATA, BUT VALUES MAY NOT ALL BE FROM THE SAME PERIOD.



Data growth

Approximately 402.74 Million TeraBytes Of Data Are Created Each Day

It includes the newly generated, captured, copied, or consumed data.

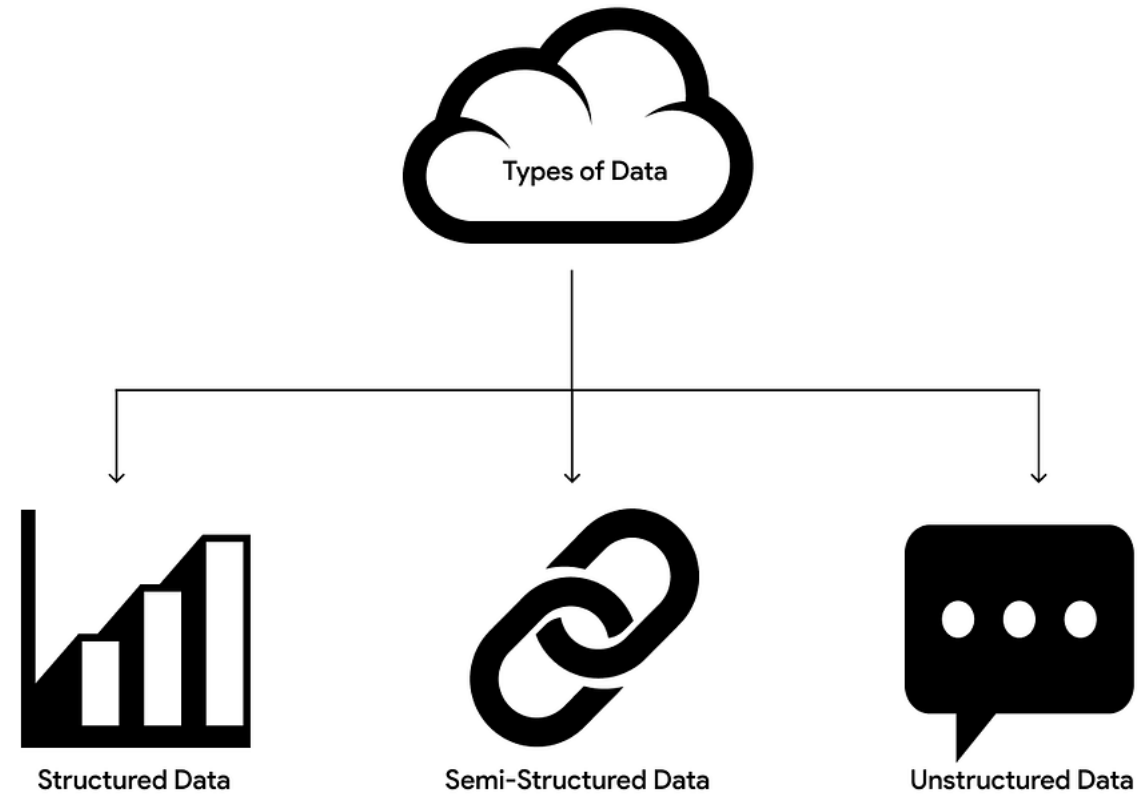
This means 90% of the world's data has been created in the last two years.

Converting the numbers into gigabytes, we discovered that 402.74 billion gigabytes of data are created daily, and 147 zettabytes of data are created in a year.

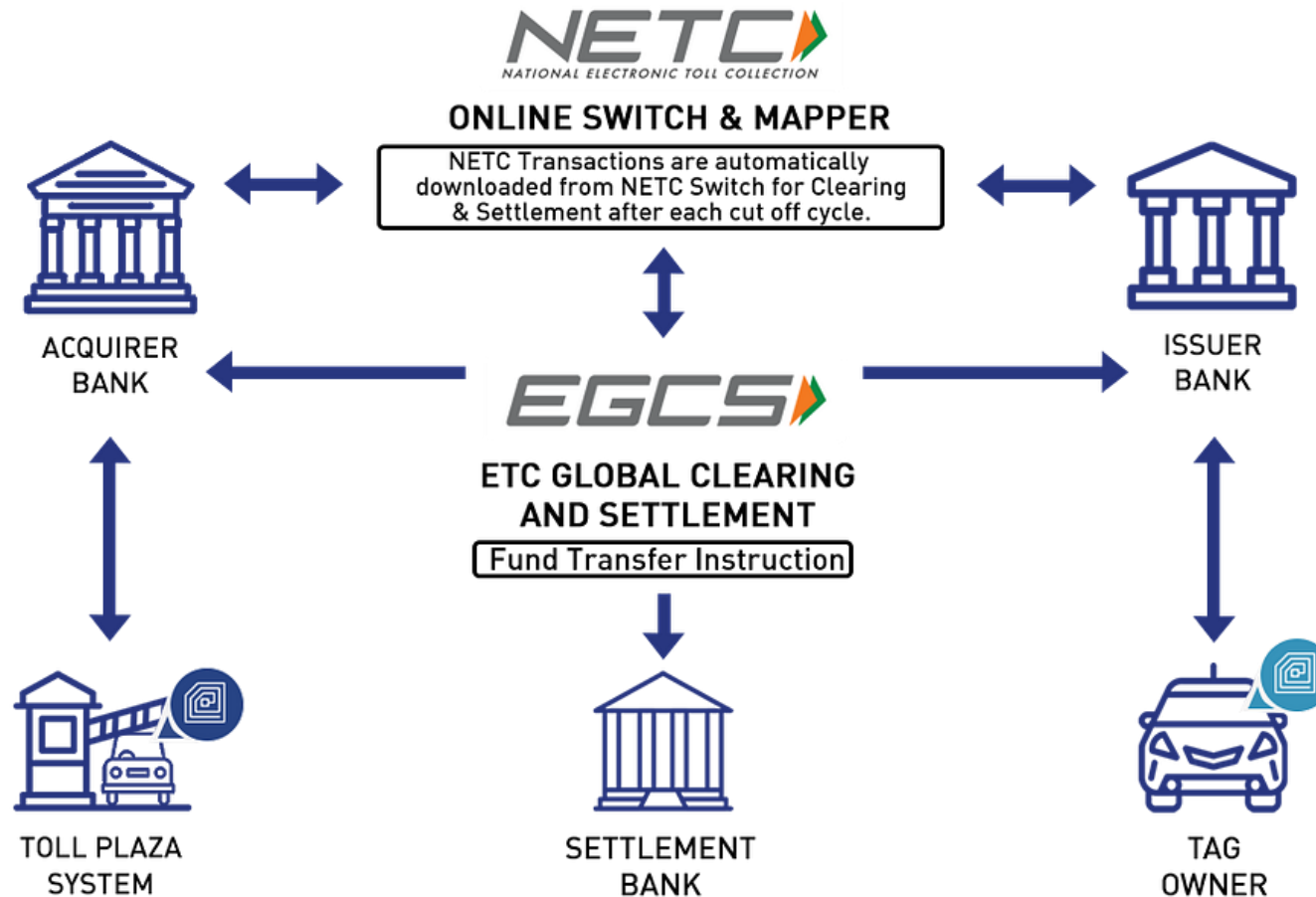
Here are further details about the amount of data created in different time spans.

Time Span	Amount Of Data Created
Per Day	402,740,000 TB
Per Hour	16,780,833 TB
Per Minute	279,681 TB
Per Second	4,661 TB

Types of Data



Data generation every minute



Data generation every minute

Semi-structured Data



References

[\[1\] Social Media Usage Statistics \(2026\): Key Facts & Insights](#)

[\[2\] https://www.npci.org.in/what-we-do/netc-fastag/product-overview](https://www.npci.org.in/what-we-do/netc-fastag/product-overview)